ActInf Textbook Group ~ Cohort 5 ~ Session 3 (Chapter 1, Part 2) ~ 10/3/20

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 3 (Chapter 1, Part 2) ~ 10/3/2023 | 59:02

all right hello welcome everyone it's
October 3rd 23 and we're in cohort 5 in
our second discussion of chapter
one we'll jump over to the uh cohort
five
questions it's not too late to add more
just please feel free to copy in
anything that you'd like us to explore
but before we get to any of the
questions is there just
any kind of top of
Mind piece about chapter one or some
specific passage or some figure or
something that anybody wants to share
about chapter one before we jump into
the question
specifically
okay
okay let's go into the questions we
we'll just go through them and again you
can continue to add more or upvote them
as as we continue
question about modeling as action it
seems that
modeling is still abstract and based on
observing and making effort to
understand Dynamics Dimensions behaviors
of given systems I.E modeling still
seems theoretical possibly distinct from
applying
models and interacting with systems in
the
wild of course observing and documenting
are also actions however the systems and
behaviors being observed would be the

primary actors expressing a rich complex
range of acting in cohort 4 chapter 1
discussion arose a question about
observing versus acting I guess it's
well documented there inextricably
intertwined what are anyone's thoughts
on
this
for me
um the perception action Loop seems to
be more in line with you how we're
making um policy
decisions and not necessarily at this
point about how we're mapping how we're
moving um and so you I don't know if
that's not you if if we can if we can
kind of avoid conflating those
two just to get the gist
of where we're at in the in the book for
me seems to
be
Cent thank you um
Jesse I'm still on the phone for like
the last day
um yeah that that reminds me of like
Maxwell's demon you know in a certain
respect where even through making those
choices you know it's it's an action so
even the generating of the model and
making that decision it's
work that's also very um interesting
point um it seems like the the
questioner recognizes that um making a
model is something that a person can do
like a a researcher or scientist may do
it
so if you're thinking from the
perspective of the modeler of course

modeling is an action um applying a
model that could be both abstracted and
still an action I mean you could
abstract away from the rat in the teamas
like an example that will come back to
or a frog jumping out of the hands which
we'll come back to you can abstract that
situation and model it and then you
could apply that model but once you're
in the map as opposed to the territory
then there might be things that the
territory could do like the rat could
jump out of the maze but your model
might not reflect that um and this is
also a very rich question about
observing versus
acting we can revisit but action is not
just overt gross bodily
movement Act
is conceived of very
broadly uh in arbitrary spaces and so
there's several ways that even in a
situation that might be seen as a
passive a passive observational setting
there's several ways which action comes
into play so visually
icting is a a situation that's been
modeled a lot in active inference so
looking through a visual
scene the is still an active policy
selection ongoing about where the eyes
move but that's still bodily in a sense
but even more
cognitively perception um is augmented
by attention and attention in active
inference is modeled by an internal
policy decision so kind of like where to
shine the internal Spotlight or how

diffuse or how Focus to shine the
attention Spotlight that's modeled as an
internal or Co covert
policy so there's definitely a lot of
ways in which action and perception
intertwine through through the loop and
otherwise and um this is ironically or
not a theoretical question about
modeling being theoretical or not
whereas once somebody puts something on
paper and gets into the process of
iterated
modeling
then it just becomes resolved in the
actual what it is that they are doing
and once you put the model on a piece of
paper or computer it's not uh it still
may be theoretical but but it it is also
embodied so so just to clar to clarify
something restraint would be considered
an action is that
correct
restraint you're not taking action
choosing to not take action that's a
that's a great question there are
several um ways in which uh timing your
take is modeled
often uh kind of like a null action is
provided as an affordance so the rat is
in the te-as they can go up down left
right or
stay and then Through Time it might
select staying so it's like selecting an
action but it's kind of like a null
action so in one conversational sense
you could say it didn't do anything but
because the model must update every time
step you have to provide an

affordance that that does
nothing so the
selection is the action go
ahead doesn't um a null action really
make more sense there's there's two ways
that null action can be it's not it's
either not modeled or it's uh
intentionally
um nked and so once you get into nested
models null action makes uh more sense
as an
intentional negation of
action so where would indecision lie or
would that just be yeah how would you
frame indecision in that
context O Okay so first to the um um
let's say we're writing out a number so
this is our our our act our action space
writing out a number that's you know
we're writing out our favorite number we
expect and prefer it to be you know this
number that we love and um this could be
modeled as a single level model so then
the affordances would be all the numbers
that you could write down or it could be
like a nested model where we're going to
write down like the hundred's place and
then the T's Place and the ones place
and so like all the decisions you can
make about the ones place are nested
within the T's Place um so that's that's
the kind of nested architectures or you
could make it a single layer model to to
um reference Andrew's point in decision
um and then let's just say we include we
have one model where you can write down
zero through nine those are so every
click of the model it must decide zero

through
nine another model you could say zero
through n or
pause and so that model depending on how
the parameters are Set uh set so
essentially how the um cognitive
diversity is in that moment maybe one
agent just Waits Waits Waits Waits Waits
Waits Waits another agent writes down a
three in the hundreds place and then
Waits um
indecision you could kind of un unpack
that in a few different ways One agent
might have very high Precision that they
know that they want to wait so that
would be indecisive in a way about what
number it would be but they would be
highly precise about what policy they
wanted to engage in another flavor of
indecisiveness would be if the um
different
options different affordances were kind
of equally or
similarly
Salient so that kind of indecisiveness
is like there's multiple options and I'm
not sure which one to
pick but the these are all these are
great comments and and it just shows how
even um behavioral settings that are
like almost too simple to to believe
like just writing down a single
digit they they give us a handle
into different cognitive
phenomena memory attention preference
decisiveness all these different
things so that's why simple toy
examples are used in the textbook

because even very simple behavioral
settings
can give
clear um it's like a example in a
grammar
book to just exhibit sentence structure
these are like simple sentences of
cognitive well-formed settings that help
us practice fluency in the ACT INF
anology
Francis I was thinking about
the uh way that acts and observing are
not actually
different and I think for me one of the
differences between indecision and
inaction
is in actions kind of intentional maybe
I'm waiting to learn something
but when I'm truly stuck and uh this
does happen to
me
uh in decision I feel almost like I'm
not actually perceiving anything either
I just um I'm in distraction mode I'm
not actually picking up anything that's
going to move me
forward
um
yeah very
interesting that's a um almost a uh
analytical Auto ethnography or
analytical Auto
phenomenology like entering in at our
own
experience and then digging or trying to
investigate like starting with what is
it like to dot dot dot yes yeah and then
like but what is it like to that's a

very highlevel cognitive phenomena like
a pendulum or um a simulated entity we
don't need to appeal to it having like a
higher order
self-concept so um that's the double
edge of entering in at the level of our
own experience on one hand what else
could be more tangible and immersive on
the other
hand the kind of phenomena that we might
be dealing with are not necessarily like
basil or the
atomic or the essential
phenomena but that's the fun is to then
explore how how could how could we
compose what it's like at a higher level
from and and there doesn't need to be
only one account and certainly not early
in that research question so then just
thinking divergently and building up
different ways that it could be modeled
understanding modeling as action like
scientific modeling as real action in
the world also it it just it opens many
doors it just helps see um scientific
pluralism in a new light like models are
proposed maps of territories so building
different Maps comparing different Maps
that's model selection model
comparison
cool boxes
aphorism uh all our models are wrong but
some of them are
useful exactly never been more true
Kristoff yeah I was just kind kind of
wondering like if um action in the sort
of active inference setting can be
really understood um much more

technically in in the sort of you know
thinking about this um in a sort of more
like a graphical way
in as there's some there's some sort of
State machine that you know represent
representation of the internal stat of
an agent and whenever there is any kind
of transition in
that that's essentially an action
something is happening and what is the
meaning of that action is completely
arbitrary we may um you know in our
model we may assign the meaning of that
action to be some kind of mental
transition it could be essentially as we
discussed the null action like I I do
believe that you know not doing anything
is very different from doing nothing
doing nothing it's a very it's a very
active State and it's not easy to be
doing nothing consistently it's it
requires a tremendous amount of practice
um so but it's up to us to effectively
assign the meanings to this but from the
perspective of the formalism that really
is kind of just a just just a state
transition and there's some sort of you
know effect that is being emitted by by
the model by doing that state transition
that is some some something can be
observed um from the perspective of
looking at the at the model as as it's
kind of running so when we when we're
updating to
State absolutely two ways in which we'll
see action as a variable in a graphical
model or in a in a state machine here in
more of the um particular

partition representation style action is
just as you said the the variable that's
emitted by the cognitive entity of
Interest so it's what goes back into
just like perception it's not taking on
some philosophical baggage of like being
experienced um it's just the incoming
statistical dependencies and in that
view action is the outgoing statistical
dependencies and then what we'll come to
in chapter um four is action um here
symbolized with pi for policy action is
just what
possible um affordance what what action
selected intervenes how in the causal
process of the world so yes it it it
ends up being
operationalized with a variable in a
model once it comes to this
stage you can also um generalize
learning as action right so you can say
action is extrinsically oriented or
intrinsically oriented
learning as as action let
um I just searched for sanit Smith
because
um yes various mental or cognitive
processes have been slash can be
understood as action like if you have
some belief distribution you could
understand that movement as being a
movement in a space and so hence an
action in a space and um Chris fields
and Mike Levan have talked about
intelligence as competence in navigating
arbitrary State
spaces and so none of that requires
overt bodily action this not the only

kind of action being discussed and then
once once you open up the door to
thinking about internal action using
control theory then that's been applied
to attention that's been applied to
learning as action and other cognitive
phenomena okay awesome great
question
okay about our discussion on variational
free energy as lower bound of a
posterior distribution I found a good
primer of ways of deriving evidence
lower bound
elbow is how some in in the basy in
statistics and uh statistical physics
world that's another like synonym for
variational free
energy um if anyone with a stronger
computational bound would be willing to
walk less proficient of us through the
guide it'd be greatly appreciated I'll
post a link in the notes if anyone in is
interested in learning it
comment I also posted in math learning
group math learning groups just kind of
um it's another math learning space
people can and read through it Peru
there's no like side group um there's
also currently no other meeting time but
in some Past Times people on the more uh
basic or fundamental math learning side
of the picture or more on the like rigor
and and their improving side they've
wanted to kind of specifically focus on
the math so um perhaps comments on that
question or or or um you know write more
here or write a a comment or anything if
you want to explore that more but looks

good and and we'll come to some of these
like mathematical technical things like
they will come up in the coming
chapters but it's great that that
someone's thinking also ahead about this
in chapter
one cool well um if anybody has more
questions
they can just paste
them in um this cohort 5
Section we can look to the overall
questions on chapter one and kind of
look at some of the most upvoted
questions and just kind of review
these questions that have been posed
some of them are are are totally core
great points or someone can just
raise their hand and and like please
feel free to just add any any point or
ask ask any
question um Andrew and then anyone else
yeah I just thought of a question
um I can add it too one thing I'm
curious about the whole uh active
inference Paradigm and it's Associated
Concepts Vision uh mechanics and and
whatnot
is uh whether there's
a good idea of what's really
state-ofthe-art and sort of loosely held
beliefs versus things that are really
cementing and and um
propagating uh as like pretty darn sure
this is the way we should think about um
various
phenomena so what are yeah what are the
loosely held beliefs in this space
versus the strongly held ones

great great question there's there's
um there's a lot to say on that and um I
I I hope as you know through time we
we'll we'll continue to to develop
better better answers to that
[Music]
um I'm thinking of how how to um reflect
what has been happening very rapidly in
the last few years to to address your
question I'll just give kind of one
historical note just to kind of plant
the seed there um Carl friston and
colleagues in the late
1900s working on neuroimaging and fmri
in the neuroimaging unit at UCL and so
that's the development of the
statistical parametric mapping or SPM
toolkit and so that's the most widely
used neuroimaging Sensor Fusion toolkit
that's what made Carl friston the most
cited neuroscientist at the beginning of
2000 Carl friston and colleagues begins
developing what was the free energy
principle which was this idea that there
could be a unified imperative or a
unified principle for all that cognitive
systems do unified imperative for
perception and action and in some papers
between 2006 to around
2012 there were a lot of um initial
foras as far as I understand there's no
major branch that was like totally
repudiated however there were different
equations where you know a Tilda had to
be unpacked or a term had to be added
um between around the the 2010 to
decade there were increased
applications which provided a different

kind of support or belief confirmation
which is to say it wasn't just about the
analytical formalisms like transforming
well between each other but also brought
in like a level of explaining and
predicting empirical data and uh for
example the work of Ryan Smith and
computational Psychiatry and during that
decade also there was a cohort of um
graduate students and and other
researchers like Maxwell ramstead axle
constant at all at all who were
beginning to say well free energy
principle for the brain which was like
Carl friston's very well-sited 2010
paper it's like well it's not just about
the brain we can expand this to think
about like ecosystems and and so that
kind of brought in like this multi scale
modeling angle and then over the last
few years um much of which you can find
reflected in the live
streams some of the
specific angles that connected to
physics which is now known as basian
mechanics but basian mechanics was not a
term used in this way several years ago
but basian mechanics
brought in a new le level of of rigor
and
grounding and it recontextualized some
of those
formalisms that had been proposed going
back to 2006 and
everything it recast those in terms of
the um the generalization Direction in
physics and put basian mechanics as a
mechanics along which but not not a

reductionistic or or um material account
but a mechanics of basian systems and
put that
alongside Newtonian mechanics Thermo
informational and quantum
mechanics live stream
49 with Dalton Sak divid devel has some
of that discussion and on the quantum
side live stream 40 series with with
Chris fields and also Chris fields's
ongoing Physics
course those represent some of the most
ongoing
developments in um re understanding free
energy Principle as
principle active inference as
implementation approach and then basian
mechanics as how that
happens and again that didn't
necessarily repudiate
earlier work it helped understand its um
scope
though and then in more targeted areas
people occasionally do write sort of
review
pieces which we can talk about but um
you know the state of the field is is in
some ways what people say here and in
the literature that's
assembled okay yeah so um to synthesize
what you're saying um essentially the
the newer research
is uh
reconceptualizing and
clarifying earlier Stu
um and so so helping put it on more
solid epistemic
foundations so so really the new stuff

is
pretty uh there's a lot of uh truth
behind it it's not um sort of tentative
exploration in terms of like the beian
mechanics seems to really getting be
getting getting a foothold in the
physics world and
um rather than being sort of
tentative I would I would agree with
that I think that's that's for each okay
person to evaluate I'll just point to
this 2010 F the 2010 first in paper that
I um mentioned there this is like a very
provocative
early positioning with like the free
energy principle and then
understanding how out of the free energy
principle potentially various special
case cognitive
phenomena but still vast and Broad
phenomena like model selection and
evolution could be described um as a
special case of free energy
principle
um in the setting of The Action
perception Loop now there's been
subsequent developments on like what do
we mean by the action perception Loop um
what about the arrow that can that can
plausibly connect you know across the
blanket like so some details have been
coming into better focus but but but
also it's only been
like months to low numbers of years that
tends to potentially only small orders
of magnitude more numbers of people
having even been able to reckon with the
quantum formalism or basy mechanics or G

Theory
so ongoing
Susan um to follow up on Andrew's
question yeah I I'll start
with uh reading kind of the summary of
chapter two which I thought we were
going to be going every chapter two but
um but it it offers kind of a
generalized um theory of active
inference is a theory of how living
artifacts and I you I I kind of put
there organisms underwrite their
existence by minimizing surprise
variational free energy via perception
and action and what I take
away um
from you know the more I read about it
which I which I like because so much of
psychology tends to be you know in
decision management it tends to be
prescriptive but taken out of
context it's more wrong than right so
what I kind of like about the um what
I'm reading is that it's not trying to
be prescriptives is trying to give us a
framework to uh to
explore how
individuals can build models and how we
can build models that are more closely
resembles
reality so for what that's
worth that's a great Point um thank you
raore I I love this historical element
and I'll just point to one more um kind
of piece of recent history um in 2019
Carl friston wrote this long monograph a
free energy principle for a particular
physics particular is a little bit of a

pun double on Tandra it's a specific
physics like if you want to make a
different physics you can people there's
Newtonian physics there's relativistic
there's all these different physics and
so this is just one specific physics so
if you have some qual with it you can
version and modify it or you can develop
a different physics but also it's not
just a specific physics or a peculiar
one it's about a particle it's about the
cognitive particle or just like the
generalized particle and so it was this
paper in
2019 that really signaled an inflection
point with like we're going to be
talking about the particle which is like
the blanket and the internal States
that's going to be like figure and it's
going to be partitioned off from ground
or the niche or the environments so it's
like if that partition can't be made
between the system of
Interest blanket and internal States the
particular States
here if that can't be partitioned off
from the niche you don't have a noun or
a thing to point at that's like almost
tological but like but but um if you can
make a distinction there's a difference
and so that's a thing and that's going
to be the particular States and then
there's some some subpartitioning of of
certain um pieces that we might call our
attention to or develop formalisms
around but it was in 2019 and then it
was a reading group with Maxwell and
others that had that brought a a

relatively lot of attention to um the
physics here and then that um that was
kind of incubated in the 2019 era led to
Dalton and
others
really um
strengthening the
physics side and
connections
Kristoff um you know I think there's a
interesting development happening right
now in the uh in that space you know
like you mentioned that uh if you can't
draw a boundary around the thing if you
can't if you can't make that blanket
then you don't really have a thing and I
think some some work that is yet to be
published that Maxwell ramstead uh
mentioned a couple times on various
podcasts recently they're
formalizing the notion of essentially a
fuzzy blanket where there is some kind
of like index into the blanket blanket n
of of of the blanket like how how how
fuzzy it really is and uh you know my uh
my prediction is that it's going to it's
going to uh turn out to be something
along the lines of that that that the
distinction is always somewhat arbitrary
that at a certain scale you have a thing
in a certain scale there is no thing you
know almost like the the Buddhist
concept of no self is going to show up
in the mathematics of it that depending
how you look at the system the internal
States really are a thing and and and
and from from a certain perspective and
from a different perspective that thing

just simply disappears if you if you
start really looking for it it's like
when you start start to look for the
boundary of the um I don't know a cell
you're going to find that that boundary
is becoming fuzzier and fuzzier the
closer you look right uh the the the
more Jagged and fractal and and and
harder to Define what what exactly is
the internal state from from the from
from the external state so um it's a
it's an interesting it's very
interesting to see how that stuff
developed you know like when in the I
think the there were a lot of ideas how
to draw that boundary very it was just
kind of like a hard boundary that you
have to have this distinction uh that
has to be very uh very strong between
the internal States and external States
and now it's slowly dissolving as the
mathematics develops and I think this
reflects the fact that mathematic is
ultimately mathematics is ultimately an
intuitive discipline um you know this
principle the principle is true and you
can you can sort of see it for yourself
but actually properly formalizing the
mathematics in such a way that it's
actually airtight in in all cases that
always takes a long long time you know
calculus was full of holes for I don't
know like a long long time like probably
until uh
um uh until um Whitehead and uh what's
this the I forgot the I forgot the other
guy princip of Mathematica came along
and started formalizing things and and

so and so on and I think the same process kind of has has been happening in the um in in the free energy principle space as well that a lot of these ideas started started off a little bit um you know based on certain mathematical assumptions that now have been revised like er ergodicity used to be somewhere there and I think now they they they they refor reformulated some of the ideas without it um and it's becoming Much More Much More airtight and that's just a natural progression great points thank you Kristoff yeah just a few Point uh pie on that um here's a 2022 uh paper um perhaps not the the only or the total work but the blanket index was proposed which is kind of like a zero to one Continuum about like fuzzy or dynamical or wandering blankets but being able to um recognize from empirical data and not have the particular partition be in a priori that then dictates the subsequent analysis but enable the discovery of things to to be um part of of the analysis process that's those are great and then also um especially with Chris fields's work on the quantum information sciences and Mike Len's work on poly Computing there is also the um increased focused on like The Observer playing a fundamental role like you can't take a view from nowhere on what computation or what cognition something is performing if it's encrypted to you it's going to look like noise if it's not encrypted you might understand what it is and then

exactly like what you said about the
spatial and temporal scale like if your
time scale is uh 100 years
then an eoli lives and dies as as a
statistical noise and you you would not
be able to perceive that as a thing
whereas if it was um you know a Pico
second time interval you wouldn't
perceive action occurring and so 100%
it's about like zooming in zooming out
and finding Salient spatial and temporal
scales of analysis which comes back to
the modeling as action there's just no
model without a modeler and there's no
observed without the
Observer and and so some of these points
are and and and then you pointed to to
some more technical discussions on
ergodicity and and requirements around
that but these um um we're reading the
first textbook of active inference which
realistically is not so much longer than
for example step by step it's it is
different but um this is not a massive
textbook however what sanj namjoshi is
doing is incredibly comprehensive and
may um his textbook coming out within
the coming you know 12 to 18 months or
whatever it is that will speak to this
question in a way that that the textbook
that we're discussing
here doesn't um lay it on so thick but
um yeah these are these are epic
discussions and and this is just like
being in the game to even enter the
slipstream to try to understand these
and also like to understand where it's
like okay erodic it's just like all

right it is that that is how it is and
people are working on those questions um
Susan
yeah it's definitely upending um a lot
of of Economic and other affable um
models that uh just don't suit the world
anymore um I'm curious um to explore uh
with what you mean um kristo am I saying
that correctly Kristoff when you're when
you're talking about
airtight so when you talk about airtight
in mathematics can you expand on that
just a slight just a
bit um so let me let me give you an
example um from mathematics that's kind
of vivid from my education the first
time I encountered uh for your analysis
you know it's like it's it's a beautiful
piece of mathematics but there's a
problem with it it just does not
converge at the ends of the intervals in
in in a lot of really really simple
cases and it took quite a lot of work to
actually make that work mathematically
right you have to um get into schwarz's
family of rapidly decreasing functions
and um generalize notion of functions
distributions and so on so forth and and
bring bringing that in incredible
mathematical rigor um you can actually
make um for air transform air tight so
to speak that you can finally really uh
stop making the handwavy argument that
oh you know like I have this limit of
some kind of gaussian um gaussian
function that in the limit actually
approaches the direct Delta
but there is there's a precise way of

defining that in terms of sches rapidly
decreasing functions and so on so forth
um and then you actually get a
formalism which you act you you get
proper results all the time but it never
really stopped anybody from making
tremendous use of um f Transformers it
was right like with these little holes
which just knew they were there um
Engineers would be aware of them and
that that that that never really stopped
anybody uh it's it's validity you know
the the validity of the idea was not um
was not questioned just because there
were certain cases in which uh it would
break down and especially that intuition
Behind these these sort of limits and
calculus is another good example I think
um you know Russell and Whitehead they
they got to um formalizing mathematics
precisely because they were worried
about this you know this this notion of
a limit in Calculus if I if I remember
my history correctly uh what what do you
mean when you know
something this this um something tends
to be it tends tends to become
infinitely small what what what does it
actually mean precisely like it it's
always been kind of like a hwa handy
argument um but it never stopped
mathematics from just working right like
there is a there is a certain
um geometry to it that you can just
simply see and symbolically in in some
sense non-constructive
mathematics is is always like that
unless you have a completely

constructive system computational system
effectively and and you derive something
from from an axotic Foundation then your
your mathematics is is is quite likely
going to be full of full of some holes
and you can you you can get into some
really interesting philosophical
arguments like what why is it even
possible that we can do something like
that to without having this precise
computation to jump over the threshold
and say like well you know I kind of I
have this intuition that this is
actually going to work out and it does
and it's it's a it's a fascinating
question but that's uh I think this is
the kind of process that we're what
we're seeing is this kind of thing where
we started off with intuitive ideas that
came from a lot of um experience of
applying mathematics in in in in in in
in this kind of setting
um then various attempts to formalize
that using the mathematical tools that
were available um at the time finding
certain cases where these tools were
insufficient and you know rederiving
this for this this formalism again and
again until until it becomes um uh you
know
uh again like I'll use the same term or
tip I hope I hope that makes sense yeah
absolutely actually I'm very sensitive
to what Andrew brought up in terms of
you know what is the underlying beliefs
I I actually you know
constantly um I'm trying to understand
ideologies because they're they're

underneath beliefs and in fact one of
the statements I wrote down from chapter
two was which was you know an assumption
was whether we adjust our beliefs or our
data depends on the confidence with
which we hold the beliefs I I think that
kind of speaks to what you're saying in
terms of confidence levels um and um you
know and and so you know when we kind of
unpack the you the surprise the
variational surprise and consider that
that is really about
uncertainty if we can you build that
mathematical model that um that more
accurately um aligns with reality then
um whether that be on an individual
basis or group basis or basis doesn't
matter I mean we're going to start
getting better results so so yeah that's
that's a great uh explanation of math
being a non-math person thank you yeah
th this was great I I love this point
like like we might be able to have an
intuition like okay what if you take a a
gaussian so like a a bell curb
distribution and you just make it
sharper and sharper and sharper and
sharper it's like can we just say that
it's a spike it's like well maybe not
necessarily but in a given application
that might work perfectly well and then
those like sort of intuition looking
across the Grand
Canyon you know could we jump across it
could we build a bridge or is this like
an incross Gap th those are the research
um adjusting our belief or the the data
this is in reference to the two ways

that free energy can be minimized so the
two ways in which surprisal can be
bounded which is change the world
through action or change the Mind
through updating the cognitive model
which might be considered learning or
perception or attention depending on on
which way so here's sensory data coming
in we don't want to have our thumb on
the sensory scale like that's kind of
like a trivial and ultimately um less
effective way to let's just say um weigh
the amount that we want to weigh well
yeah there's a trivial way to make that
happen which is just change the number
on the scale so we take the thumb off
the scale sensory data we don't directly
control what are the two ways that the
particular State can then come
into uh come to bound its surprise or
minimize its surprise change your mind
change the internal state
or change the world through action those
are called the autonomous States just
the internal and the active States so
that's an example of like a subset of
the particular States it's two out of
the three particular States but then we
can talk about particular states to
differentiate the thing from its Niche
so that's these
three and then you might want to talk
about just the blanket States just the
the two in the middle perception in
action or you might want to talk about
the autonomous States just action and
internal States because those are the
controllable ones in an optimal setting

if you can control what to think and
control it to do and then your actions
through the niche end up influencing
your observations and hopefully linning
them with your preferences that's
pragmatic
value
cool and figure uh one and
two figure one one kind of grounds Us in
fundamental agent World
interaction so this is even simpler than
the four-fold partition that we see here
this is just kind of the two fold
partition just figure from ground or
like agent from Niche agent-based
modeling and then figure one
two lays out quite literally like the
road map on how we're going to get to
active inference in the coming chapters
the low road is a how question it starts
with basian Statistics and it's going to
work up towards basian cognitive model
models but basian cognitive models could
be of any kind you could interpret a
program as a basian cognitive model or
there's nothing about an arbitrarily
constructed basian model that makes it
like adaptive in the niche but that's
the how and then coming from the high
road which is chapter three we have this
persistance or repeated measurements
imperative things that can't be
repeatedly measured
Meed are going to be understood to be
noise and not signal they're just not
persistent long enough to be measured so
as an observer it isn't a thing in our
data if it's changing at a time scale

faster than our um like persistence can
understand or slower um and then when
that gets turned back on the agent
itself as long as you're remeasuring
your body temperature to be in a
homeostatic range you're alive and so
that's
how the um survival or self-organization
or self-
evidencing is understood but this
doesn't uniquely suggest a how there
might be different ways to develop
self-organized self-
evidencing surprise minimizing marov
blanketed
systems but the RO map of this
textbook is that in Chapter 2 we're
going to learn about the how and about
how basian statistics and
updating can be used to develop models
of perception and
action and then you know
refresh how systems can be
understood as embodying this principle
of persistence because we're talking
about systems that do
exist even if they're only proposed to
exist and then those two roads come
together in active inference and then
right in chapter four we're going to be
talking about the generative models of
active inference which use this how for
why and there they
are um yeah in the last minutes anyone
have any other thoughts on one or about
like where we're heading in the coming
weeks I do have one question about um
kind of tying into my earlier comment on

beijan
mechanics do you have a feel of
what um whether we touch on the
connection mentioned in basan mechanics
between um I I believe it's called
entropy maximization as like the
external View and I think the textbook
is mostly this sort of internal view but
whether we touch on that other piece at
all great comment so um the work of
Dalton and Maxwell and others um in
check out live stream
49 um or the guest stream with uh with
Dalton let's see what number that
was guest stream 23 rebooting the free
energy principal
literature in this work as you alluded
to it came to be understood and
demonstrated that the free energy
principle is like the view from the
outside on persistence and then the
constrained maximum entropy principle
CME is like the view from the obs
Observer which is like we should kind of
be like maximally agnostic about what
that thing is going to do but but no
more agnostic than that and then um the
view from the inside um is the free
energy principle and they unpack this in
work um it's
not heavily addressed in this textbook
as it was more of a recently
breaking and more technical point but
these discussions help connect Fe with
CME cool
well um people can can make any final
comments otherwise this was an awesome
discussion on

one and then in the coming two weeks we'll be heading into the low road with chapter too so in the last minutes does anyone have any other comments that they want to make just about heading out of chapter one or heading into chapter two uh I have one like there there's something interesting that I run into recently that I I think to me it's it's very much related to active inference uh but it's um it's it's it's a bit of a sort of a uh why they suppose relationship I I recently run into a bunch of interesting literature on the subject of um inside meditation so um essentially Buddhist Traditions from the terada tradition or more recent stuff like mahasi sa's um Manual of inside and so on and ultimately uh you know everything that that that Gama Buddha uh was was was trying to teach people and like all these all these Med all all these meditation teachers uh that um came after him are are trying to teach is effectively a method for um building a generative model of the Mind from the inside and it's extraordinarily simple it it really is about paying attention to what's happening in the mind and then having a vocabulary to label what's what's going on I am thinking I am intending I am looking I am moving I am feeling this I am I am I am thirsty I am cold I am you know uh feeling pressure or uh pain or or pleasure and so on so forth and the entire um system is based

around literally noticing that this is
what's happening and constructing ever
so ever more sophisticated models of
what's actually happening with the mind
um and interestingly like the only the
only thing that there is for a person to
do um to acquire all these skills is
just observation because the the basic
premise is that there is no there is
nothing you can do
to make your mind do anything it will do
things on its own and all you can do is
to kind of notice that this is what has
happened and then hold an intention that
this is what you would like to happen
more and if you hold that intention and
you sort of project positive energy
towards the outcomes you want over time
uh this reinforces the the the the the
the um you know occurrence of these
events uh more often so you become more
mindful the mental states that you want
to cultivate increase the mental states
that you um that are UNH wholesome
diminish and and so on so forth but it's
really fascinating if you start thinking
about this in terms of literally
constructing a generative model that I
have a model of what my mind might be
doing and in every moment of perception
when something is happening it hits the
model and I can and I can see like oh
these are you know of all this of all
these internal of all these states that
I have mapped out this is um this is
this is what's happening here so this
relationship between you know the
mathematical Theory and something that

you can actually phenomenologically investigate by yourself by you know just sitting quietly and and and looking at what what your mind is doing is is really

fascinating thank you we're at the end of our time so I will close it but awesome discussion thank you cohort 5 and looking forward to seeing you in the coming weeks and please add questions add writings you know if this Sparks something you just just add questions and and insights and we will um go from there so thank you and farewell see you everybody bye